MODE STABILITY FOR GRAVITATIONAL PERTURBATIONS OF SUBEXTREMAL KERR–DE SITTER BLACK HOLES

JAIDEEP SAI PADHI

Abstract. We prove mode stability for the Teukolsky equations of spin $s = \pm 2$ on Kerr–de Sitter black hole spacetimes in the full subextremal range: there are no nontrivial outgoing mode solutions at frequencies σ with $\operatorname{Im}\sigma \geq 0$. We then show that every generalized mode solution of the ungauged linearized Einstein equation $D_g\mathrm{Ric}(h) = \Lambda h$ with $\operatorname{Im}\sigma \geq 0$ is pure gauge if $\sigma \neq 0$, and the sum of a linearized Kerr–de Sitter metric and a pure gauge term if $\sigma = 0$. This is the mode stability hypothesis under which Hintz, Petersen and Vasy proved the nonlinear stability of Kerr–de Sitter black holes in the full subextremal range. For real frequencies, the Teukolsky part follows Hintz's two-current method for scalar waves, applied to a dual radial equation obtained from an Euler integral transform. The upper half plane is reached by a continuity argument in which high-frequency control is borrowed from the gauge-fixed Einstein operator through metric reconstruction. The passage from the Teukolsky scalars to the metric uses a linearized Goldberg–Sachs argument and the two Killing fields built from the Killing spinor. Several steps are computer-assisted, using exact arithmetic.

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1. Introduction

For $\Lambda > 0$, $M > 0$ and $a \in \mathbb{R}$, the Kerr–de Sitter (KdS) metric in Boyer–Lindquist coordinates is

$$g = -\frac{\Delta_r}{\Xi^2\varrho^2}\big(dt - a\sin^2\theta\, d\varphi\big)^2 + \varrho^2\Big(\frac{dr^2}{\Delta_r} + \frac{d\theta^2}{\Delta_\theta}\Big) + \frac{\Delta_\theta \sin^2\theta}{\Xi^2\varrho^2}\big((r^2+a^2)\, d\varphi - a\, dt\big)^2, \tag{1.1}$$

where

$$\Delta_r = (r^2{+}a^2)\Big(1{-}\frac{\Lambda r^2}{3}\Big){-}2Mr, \quad \varrho^2 = r^2{+}a^2\cos^2\theta, \quad \Delta_\theta = 1{+}\alpha\cos^2\theta, \quad \alpha = \frac{\Lambda a^2}{3}, \quad \Xi = 1{+}\alpha.$$

It solves $\mathrm{Ric}(g) = \Lambda g$. The parameters are *subextremal* if Δ_r has four distinct real roots, which we label

$$r_3 < 0 \le r_0 < r_1 < r_2,$$

where $r_0 = r_C$ is the Cauchy horizon, $r_1 = r_e$ the event horizon and $r_2 = r_c$ the cosmological horizon; then $r_0 + r_1 + r_2 + r_3 = 0$. The domain of outer communications is $r_1 < r < r_2$. We use a time function t_* and an azimuthal angle φ_* that are smooth across the future event and cosmological horizons, as in [4, 3]; a *mode* at frequency $\sigma \in \mathbb{C}$ is a function, or a tensor, of the form $e^{-i\sigma t_*}v$ with v independent of t_* and smooth on $[r_1 - \delta, r_2 + \delta] \times S^2$. In the language of [3], such modes are *outgoing*.

The nonlinear stability of the KdS family was proved by Hintz and Vasy [5] for very slowly rotating black holes. Hintz, Petersen and Vasy [4] extended this to the full subextremal range, conditional on mode stability for the ungauged linearized Einstein equation [4, Assumption 1.2]. Hintz [3] recently proved mode stability for scalar waves in the full subextremal range, and left the Teukolsky equations of spin $s = \pm 2$ open. Our main results settle both questions.

Theorem 1.1 (Teukolsky mode stability)**.** *On every subextremal KdS spacetime, the Teukolsky equations of spin $s = +2$ and $s = -2$ have no nontrivial outgoing mode solutions at any frequency $\sigma \in \mathbb{C}$ with $\operatorname{Im}\sigma \ge 0$, including $\sigma = 0$. The same holds for generalized modes $\sum_{j<k} t_*^j e^{-i\sigma t_*} v_j$.*

Theorem 1.2 (Ungauged linearized Einstein mode stability)**.** *Let (Λ, M, a) be subextremal with $a \neq 0$, and let $T = \partial_{t_*} + \frac{a}{r_2^2+a^2}\partial_{\varphi_*}$. Let h be a smooth symmetric 2-tensor with $D_g\mathrm{Ric}(h) = \Lambda h$.*

(1) If $(\mathcal{L}_T + i\sigma)h = 0$ with $\sigma \neq 0$ and $\operatorname{Im}\sigma \ge 0$, then $h = \mathcal{L}_X g$ for a smooth vector field X.
(2) If $(\mathcal{L}_T)^k h = 0$ for some $k \ge 1$, then $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$ with $c_M, c_a \in \mathbb{C}$ and X smooth.

For $a = 0$, Theorem 1.2 is [5, §7]. Theorem 1.2 is [4, Assumption 1.2] (with $\mathcal{L}_X g = 2\delta^* X^\flat$), so [4, Theorem 1.4] now applies unconditionally:

Corollary 1.3. *Subextremal Kerr–de Sitter black holes are nonlinearly stable as solutions of the Einstein vacuum equations with positive cosmological constant, in the sense of* [4, Theorem 1.4].

1.1. **Strategy.** The proof of Theorem 1.1 has three parts.

Real frequencies (§§3–4). After separation of variables, outside the superradiant range a Wronskian identity suffices. Inside it, we do not work with the radial Teukolsky equation itself: we apply an Euler (Riemann–Liouville) integral transform to its Heun form. The transform is injective, and its image solves a dual Heun equation that is *real*. Its potential has the very simple form

$$L(z) = \frac{A}{z} + B - 3z, \qquad A > 0,$$

in a variable z in which the event and cosmological horizons sit at $z = 1$ and $z = t$. The dual equation has exactly the structure of the scalar radial equation in [3], so Hintz's two-current method applies. It reduces mode stability to an *event lemma*: an ordinary differential inequality in the four parameters (t, z_0, A, B). A lower bound on the spin-weighted spheroidal eigenvalue gives the a priori bound $B \geq 5.5t - 1.25$. We prove the event lemma in three regimes. In the hardest one we follow [3, §4.3.3]: we first saturate one of the two conditions and then solve the other with equality. The remaining claims reduce to sign statements about explicit polynomials, which we certify by exact Bernstein-basis branch and bound.

Zero frequency (§5). This is an energy identity combined with a Teukolsky–Starobinsky map.

The upper half plane (§6). We argue by continuity along a path of subextremal parameters ending at Schwarzschild–de Sitter (SdS), as in [3, §6]. The new difficulty compared with scalar waves is that no high-frequency estimate for the Teukolsky operator in the full subextremal range is available. We circumvent this.

- Using a Teukolsky mode as an outgoing-radiation-gauge Hertz potential [14] produces a linearized Einstein mode with nonvanishing Ψ_4.
- Gauge fixing turns it into a quasinormal mode of the gauge-fixed operator of [4].
- The high-energy estimates of [4] then confine such modes to a compact set.

Nonvanishing of Ψ_4 uses an elementary observation: away from an explicit discrete set of frequencies, nothing annihilated by $(\mathfrak{p}')^4$ can be regular at *both* horizons. This uses the cosmological horizon in an essential way.

Theorem 1.2 (§7). Once the Teukolsky scalars vanish, a linearized Goldberg–Sachs argument shows that the perturbed spacetime is algebraically special to first order. Two Killing fields can then be built covariantly from the perturbed Weyl tensor: the field ξ obtained from the Killing spinor, and Carter's field ζ. Both remain Killing to first order. Moreover, $\operatorname{Im}\delta\xi$ is a Killing field of the background, so it vanishes for every mode with $(\sigma, m) \neq (0, 0)$. This makes h pure gauge, with an explicit gauge vector. The stationary axisymmetric sector reduces to Carter's separable ansatz, on which the field equations are linear.

Computer assistance. Some steps rely on exact symbolic computation or on certificates in exact rational arithmetic. They are listed in Appendix A. No step relies on floating-point evidence.

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2. Preliminaries

2.1. **Parameters.** By Vieta's formulas,

$$\sum_j r_j = 0, \qquad \sum_{i<j} r_i r_j = -\frac{3(1-\alpha)}{\Lambda}, \qquad r_0 r_1 r_2 r_3 = -\frac{3a^2}{\Lambda}. \tag{2.1}$$

We write $\rho_j = r_j^2 + a^2$, $\Omega_j = a/\rho_j$, $D_j = |\Delta_r'(r_j)|$ and $P_j = \rho_j/D_j$. The signs of $\Delta_r'(r_j)$ are $(-,+,-,+)$ for $j = 0,1,2,3$. Following [3, §6, footnote 5], we use the roots as parameters.

Lemma 2.1 (Rotation bound)**.** *On every subextremal KdS spacetime, $\alpha < 7 - 4\sqrt{3}$.*

Proof. By scaling we may assume $\Lambda = 3$. Then $-\Delta_r = r^4 - (1-\alpha)r^2 + 2Mr - \alpha$, so $p_2 := \sum r_j^2 = 2(1-\alpha)$ and $\mathcal{A} := -r_0r_1r_2r_3 = \alpha$. Scaling a configuration of roots by $\tau > 0$ multiplies $\mathcal{A}$ by τ^4 and p_2 by τ^2. Hence α is the positive root of $\alpha = 1 - \frac{1}{2}\sqrt{\alpha/k}$, where $k := \mathcal{A}/p_2^2$ is scale invariant, and this root is increasing in k. Put $S = -r_3 = r_0 + r_1 + r_2$. By the AM–GM inequality, $r_0r_1r_2 \le S^3/27$, and by the QM–AM inequality, $r_0^2 + r_1^2 + r_2^2 \ge S^2/3$. Thus $k \le (S^4/27)/(16S^4/9) = 1/48$, with equality only if $r_0 = r_1 = r_2$, which subextremality excludes. At $k = 1/48$ the root is $\alpha = 12(3^{-1/2} - \frac{1}{2})^2 = 7 - 4\sqrt{3}$. □

2.2. **The Teukolsky equations and separation of variables.** For σ real, the Teukolsky equation of spin s separates as $e^{-i\sigma t + im\varphi}R(r)S(\theta)$ [8]. The angular operator is then symmetric with compact resolvent. As in [3, §3], a mode solution can therefore be decomposed into separated ones, each with an eigenvalue $\lambda \in \mathbb{R}$. We only separate variables for real σ. Put

$$K(r) = \Xi\big((r^2+a^2)\sigma - am\big), \qquad \kappa_j = \frac{iK(r_j)}{\Delta_r'(r_j)}.$$

The residue theorem applied to ρ/Δ_r, $1/\Delta_r$ and r^k/Δ_r gives

$$\sum_j \kappa_j = 0, \qquad \sum_j r_j\kappa_j = -i\Xi\sigma, \qquad \sum_j r_j^2\kappa_j = 0, \tag{2.2}$$

together with $P_1 + P_3 = P_0 + P_2$ and $D_1^{-1} + D_3^{-1} = D_0^{-1} + D_2^{-1}$.

Lemma 2.2 (Boundary conditions)**.** *A separated mode is outgoing if and only if*

$$R \in (r-r_1)^{-\kappa_1 - s}\, C^\infty([r_1, r_2)) \cap (r_2 - r)^{\kappa_2}\, C^\infty((r_1, r_2]).$$

Proof. This follows by writing $t = t_* + T(r)$ and $\varphi = \varphi_* + \Phi(r)$, with the singular parts of T and Φ at r_1 and r_2 as in [3, (3.4)]. The additional factor Δ_r^{-s} at r_1 comes from the boost relating the Kinnersley tetrad to a tetrad that is regular across the event horizon. The Kinnersley tetrad is already regular across the future cosmological horizon in outgoing coordinates. □

2.3. **Heun form.** The Möbius map z with $z(r_0, r_1, r_2, r_3) = (0, 1, t, \infty)$, where

$$t = \frac{(r_1 - r_3)(r_2 - r_0)}{(r_1 - r_0)(r_2 - r_3)} > 1,$$

turns the radial Teukolsky equation into a Heun equation with singular points $0, 1, t, \infty$. We write $\mu(z) = z(z-1)(t-z)$. At z_j $(j = 0, 1, t)$ the exponents are those of Lemma 2.2, and at ∞ they are $\{1 + 2s,\ 1 + s - 2\kappa_3\}$. Two constants will appear repeatedly:

$$C_0 = \frac{3}{\Lambda(r_1 - r_0)(r_2 - r_3)}, \qquad c_1 = 1 + t + 3(1-\alpha)C_0. \tag{2.3}$$

3. Real frequencies: the radial problem

Throughout this section $\sigma \in \mathbb{R} \setminus \{0\}$. We fix a separated outgoing mode with eigenvalue λ. By the symmetry $(\sigma, m, a) \mapsto (-\sigma, -m, a)$ we may assume $m \ge 0$.

3.1. **The superradiant window.** Set $e_1 = 2(\kappa_1 + \kappa_3)$ and $e_t = 2(\kappa_2 + \kappa_3)$. For real σ the κ_j are purely imaginary.

Proposition 3.1 (The window)**.** *Let $m > 0$ and $w = \sigma/m$. Then $\operatorname{Im} e_1 = 2\Xi m f_1(w)$ and $\operatorname{Im} e_t = 2\Xi m f_2(w)$, where*

$$f_1(w) = P_1(w - \Omega_1) + P_3(w - \Omega_3), \qquad f_2(w) = -P_2(w - \Omega_2) + P_3(w - \Omega_3).$$

The function f_1 is increasing and f_2 is decreasing, and their zeros satisfy $0 < w_2 < w_1$. Consequently $\operatorname{Im} e_1 \cdot \operatorname{Im} e_t > 0$ exactly on the open interval (w_2, w_1), the window*.*

Proof. The formulas follow from the definition of κ_j and the signs of $\Delta_r'(r_j)$.

- *Monotonicity.* The slope of f_1 is $P_1 + P_3 > 0$. By the residue identities, the slope of f_2 is $P_3 - P_2 = P_0 - P_1$. Moreover $D_0/D_1 = f(r_0)/f(r_1)$ for the concave parabola $f(x) = (x - r_3)(r_2 - x)$, whose vertex $-(r_0 + r_1)/2$ lies to the left of r_0. Hence $D_0 > D_1$, and with $\rho_1 > \rho_0$ this gives $P_1 > P_0$.
- *Positivity of w_2.* The same comparison gives $w_2 = a(D_1^{-1} - D_0^{-1})/(P_1 - P_0) > 0$.
- *Ordering $w_2 < w_1$.* Cross-multiplying, the terms involving a^2 cancel, and $D_0 D_1 D_3$ times the difference $w_1 - w_2$ factors as $2(r_0 - r_1)(r_0 - r_3)(r_1 - r_3)(r_0 r_3 - r_1 r_2)$. All four factors are negative, since $r_0 r_3 \le 0 < r_1 r_2$, so the product is positive. □

For every real $\sigma \neq 0$, Proposition 3.4 below puts the dual equation in the real form (3.3), and u is outgoing at both ends. Let B_1, B_t be the boundary coefficients of u at $z = 1, t$ in the sense of [3, (3.5)]. Conservation of $\mu \operatorname{Im}(u'\bar{u})$ gives, as in [3, §3],

$$\operatorname{Im} e_1 \, |B_1|^2 = t \, \operatorname{Im} e_t \, |B_t|^2. \tag{3.1}$$

Corollary 3.2. *There are no outgoing modes with $\sigma \in \mathbb{R} \setminus \{0\}$ outside the open window, nor any with $m = 0$.*

Proof. For $m = 0$, $\operatorname{Im} e_1 = 2\Xi\sigma(P_1 + P_3)$ and $\operatorname{Im} e_t = 2\Xi\sigma(P_3 - P_2) = -2\Xi\sigma(P_1 - P_0)$ have opposite signs. Off the closed window the same holds by Proposition 3.1, so (3.1) forces $B_1 = B_t = 0$. At the endpoints exactly one of $\operatorname{Im} e_1, \operatorname{Im} e_t$ vanishes, since $w_1 \neq w_2$, and (3.1) gives $B_t = 0$ or $B_1 = 0$ at the other end. In each case the indicial roots at that end differ by the non-integer e_t or e_1, so $u = 0$ as after [3, (3.6)]. Then $\tilde{y} = 0$, which contradicts Proposition 3.3. □

3.2. **The dual equation.**

Proposition 3.3 (Euler transform)**.** *Let $\sigma \in \mathbb{R} \setminus \{0\}$, and let $y \neq 0$ be the Heun-form radial function of an outgoing mode, normalized so that it is analytic at $z = t$ and lies in $(z-1)^{\rho_1} C^\omega$ near $z = 1$, where $\rho_1 = -2\kappa_1 - s$. Put $\nu = s - 1 - 2\kappa_3$. Then there is a solution $\tilde{y} \neq 0$ of a Heun equation $\widetilde{\mathcal{H}}\tilde{y} = 0$, obtained from y by a Riemann–Liouville integral of order $\nu + 1$ based at $z = 1$, with the following properties.*

(i) The exponents of $\widetilde{\mathcal{H}}$ at $z_j \in \{0, 1, t\}$ are 0 and $-2(\kappa_j + \kappa_3)$.
(ii) $\tilde{y}$ is analytic at $z = t$, and near $z = 1$ it lies in $(z-1)^{-e_1} C^\omega$, where $e_1 = 2(\kappa_1 + \kappa_3)$.

Proof. That the Euler transform with kernel $(z - w)^\nu$ maps solutions of a Heun equation to solutions of a Heun equation, when $\nu + 2$ is an exponent at ∞, is classical [6, 12]; the exponents in (i) follow from those at ∞ listed in §2. Write $I^{\nu+1}$ for the Riemann–Liouville integral based at $z = 1$, defined on $(z-1)^\beta C^\omega$ by analytic continuation in β; on powers it acts by

$$(z-1)^\beta \mapsto \frac{\Gamma(\beta + 1)}{\Gamma(\beta + \nu + 2)} (z-1)^{\beta + \nu + 1}. \tag{3.2}$$

For $\beta = \rho_1 + k$ we have $\beta + \nu + 2 = k + 1 - e_1$, and $e_1 \in i\mathbb{R}$, so the denominator never sits at a pole of Γ.

Generic case. If $\rho_1 \notin \{-2,-1\}$, every factor in (3.2) is finite and nonzero: for $\operatorname{Im}\kappa_1 \neq 0$ the exponent ρ_1 is non-real, and for $\kappa_1 = 0$, $s = -2$ we have $\rho_1 = 2$. Put $\tilde{y} = I^{\nu+1}y$. Then $\tilde{y} \neq 0$ and $\tilde{y} \in (z-1)^{\rho_1+\nu+1}C^\omega = (z-1)^{-e_1}C^\omega$.

Threshold case $\kappa_1 = 0$, $s = +2$, so $\rho_1 = -2$. Write $y = (z-1)^{-2}(a_0 + a_1(z-1)) + y_{\mathrm{an}}$, with y_{an} analytic at 1. If $(a_0, a_1) \neq 0$, let $\tilde{y}$ be the Pochhammer-contour version of the transform. It annihilates analytic functions and maps $(z-1)^{-2}, (z-1)^{-1}$ to nonzero multiples of $(z-1)^{\nu-1}, (z-1)^{\nu}$; here $\nu \notin \mathbb{Z}$ because $\kappa_3 \neq 0$ when $\kappa_1 = 0$. If $a_0 = a_1 = 0$, then y is analytic at 1, and we take $\tilde{y} = I^{\nu+1}y$, which is a convergent integral with nonzero factors in (3.2). In both cases $\tilde{y} \neq 0$ and $\tilde{y} \in (z-1)^{\nu-1}C^\omega = (z-1)^{-e_1}C^\omega$.

Analyticity at $z = t$. Write $\tilde{y}(z) = (z-1)^{\nu+1}\int_0^1 (1-\tau)^\nu y(1+(z-1)\tau)\,d\tau$, or the corresponding Pochhammer integral. Since y is analytic near $(1,t]$ and the singular points $0, \infty$ lie away from the contour, $\tilde{y}$ is analytic near $z = t$. Finally, $\tilde{y}$ solves $\widetilde{\mathcal{H}}\tilde{y} = 0$ because the boundary terms vanish: in the convergent case by the vanishing of the integrand at $w = 1$, and for the Pochhammer contour because it is closed. □

Proposition 3.4 (The real dual)**.** *For real* $\sigma \neq 0$*, after multiplication by an explicit nonvanishing gauge factor, the equation* $\widetilde{\mathcal{H}}\tilde{y} = 0$ *becomes*

$$-(\mu u')' - \frac{q(z)^2}{\mu}u + L(z)u = 0, \qquad q(z) = q_1(z - z_0), \qquad L(z) = \frac{A}{z} + B - 3z, \tag{3.3}$$

with real coefficients, q *affine, and* $q(1)$, $q(t)$ *proportional (with a common sign) to* $\frac{1}{2}(t-1)\operatorname{Im} e_1$ *and* $-\frac{1}{2}t(t-1)\operatorname{Im} e_t$*. If* w *lies in the open window, then moreover:*

- $q = q_1(z - z_0)$ *with* $q_1 > 0$ *and* $z_0 \in (1,t)$*, after replacing* u *by* $\bar{u}$ *and* q *by* $-q$ *if necessary;*
- $A = t\big(X_0^2 - (X_1 + X_t)^2\big)$ *with* $X_j = |\kappa_j + \kappa_3|$*, and* $A > 0$*;*
- $B = \lambda C_0 + c_1$.

The transformed solution u *is outgoing for* (3.3) *in the sense of* [3, (3.4)]*: near* $z = 1$ *it is* $(z-1)^{-iq(1)/\mu'(1)}$ *times a smooth function, and near* $z = t$ *it is* $(t-z)^{-iq(t)/|\mu'(t)|}$ *times a smooth function.*

Proof. For real σ all exponents of $\widetilde{\mathcal{H}}$ at $0, 1, t$ are purely imaginary, and the gauge factor $\prod_j (z - z_j)^{-\operatorname{Re}(\cdot)}$ removes the real parts. The resulting operator is formally real, and writing it in the form (3.3) determines $q(1)$ and $q(t)$ from the exponents at $z = 1$ and $z = t$. We take q to be linear. By Proposition 3.1, $q(1)q(t) < 0$, so q has a unique zero $z_0 \in (1,t)$, and the sign of q_1 is fixed by replacing u with $\bar{u}$ if necessary. The remaining claims are as follows.

- *Form of* L. The function $L := (q^2 - \mu^2 V)/\mu$ is then rational, and $L = n_0/z + n_1 + n_2 z$. Expanding at $z = \infty$ with the exponents $\{1 + 2s, 1 + s - 2\kappa_3\}$ gives $n_2 = -3$, since the exponent difference there is ± 4.
- *The coefficient* n_0. Expanding at $z = 0$ gives $n_0 = t(X_0^2 - (X_1 + X_t)^2)$. In the window $X_0 - X_1 - X_t = 2\operatorname{Im}\kappa_3$, so $n_0 = 2t\,\operatorname{Im}\kappa_3\,(X_0 + X_1 + X_t)$. This is positive because $\operatorname{Im}\kappa_3 > 0$ for $w > \Omega_3$, and $\Omega_3 < \Omega_2 < w$ since $|r_3| > r_2$.
- *The coefficient* n_1. The identity $n_1 = \lambda C_0 + c_1$ is an exact identity between rational functions of $(r_0, r_1, r_2, \sigma, m, \lambda, a, s)$ after elimination by (2.1) and (2.2). Its verification is computer algebra (Appendix A, item C1): the numerators of $n_1 - (\lambda C_0 + c_1)$ and $n_2 + 3$ are divisible by $s^2 - 4$.
- *Outgoing behavior.* This follows from Proposition 3.3(ii) and the matching of exponents. □

Lemma 3.5 (A priori bounds)**.** *Throughout the open window,* $L' < 0$ *on* $[1,t]$*,* $L(t) > 0$ *and* $B \geq 5.5t - 1.25$.

Proof. First, $L' = -A/z^2 - 3 < 0$.

Next, by (2.1), $2(1-\alpha)C_0 - (2t-1)$ is a positive multiple of $3(r_0+r_1)^2 \geq 0$, so $C_0 \geq \frac{2t-1}{2(1-\alpha)}$. This gives $c_1 \geq 4t - \frac{1}{2}$.

Finally, the spin-weighted spheroidal eigenvalue satisfies $\lambda \geq \frac{3}{2}$ on the window (Proposition 3.6). Hence $B = \lambda C_0 + c_1 \geq \frac{3}{2} \cdot \frac{2t-1}{2} + 4t - \frac{1}{2} = 5.5t - 1.25$, and $L(t) \geq B - 3t > 0$. □

Proposition 3.6 (Angular bound)**.** *For $s = \pm 2$ and $w = \sigma/m$ in the window, the eigenvalue satisfies $\lambda \geq \frac{3}{2}$.*

Proof. On the window $0 < w$ (Proposition 3.1) and $aw \leq 1$, since w_1 is a positive-weight average of Ω_1 and Ω_3, and each $\Omega_j < 1/a$. Thus $\xi := a\sigma$ satisfies $0 \leq m\xi \leq m^2$. Also $\alpha < 7 - 4\sqrt{3}$ by Lemma 2.1. Write the angular operator as $H = H_0 + P + \mathcal{R}$, where:

- H_0 is the operator at $\alpha = \xi = 0$, with spectrum $\{\ell(\ell+1) - s(s+1)\}$;
- P is polynomial in $\cos\theta$ with coefficients linear in ξ;
- $|\mathcal{R}| \leq C\alpha$.

Case $|m| \geq 3$. The terms linear in $m\xi$ are nonnegative, and the remaining perturbation is bounded below by $-6\alpha - 4\alpha(1+\alpha)|m|$. This gives $\lambda \geq m^2 + |m| - 4.43 - 0.31|m| \geq 6.6$.

Case $|m| \in \{1, 2\}$. We use a two-level bound. The lowest state of H_0 in the sector (m, s) is the unique $\ell = 2$ state, and the next eigenvalue exceeds it by at least 6. For a normalized trial state $c\psi_0 + \phi$ with $\phi \perp \psi_0$,

$$\langle H(c\psi_0+\phi), c\psi_0+\phi\rangle \geq (2 + \langle P\rangle - C\alpha)|c|^2 + (8 + \inf P - C\alpha)\|\phi\|^2 - 2\langle P^2\rangle_c^{1/2}|c|\|\phi\|.$$

The moments $\langle P\rangle$ and $\langle P^2\rangle$ are exact Beta integrals. The minimum over $(\alpha, \xi) \in [0, 0.0718] \times [0, |m|]$ is certified by interval arithmetic (Appendix A, item C2). The certified minima are 1.957, 1.665, 1.920 and 1.914 for $(m, s) = (1, -2), (1, 2), (2, -2), (2, 2)$; all exceed 3/2. □

3.3. **Currents.** On the dual equation (3.3) we use the currents of [3, Lemma 4.1], in the form with variable potential from [3, §7]. Let $u_\pm = \exp(\mp i \int_{z_0}^z q/\mu)\, u$; these are smooth down to $z = t$ and $z = 1$, respectively. For a real function f, define

$$J^f[v] = -\mu f q |v'|^2 - \mu f q_1 \operatorname{Re}(v'\bar{v}) + \big(Lfq + \tfrac{1}{2}q_1\mu f'\big)|v|^2,$$
$$C^f = \mu' f q - 2\mu f q_1 - \mu f' q, \qquad D^f = q(fL)' + \tfrac{1}{2}q_1(\mu f')'.$$

Then $\frac{d}{dz}J^f[u_\pm] = C^f|u_\pm'|^2 + D^f|u_\pm|^2$. The proof is the integration by parts of [3, Lemma 4.1], using $q' = q_1$.

Lemma 3.7 (Horizon inequality)**.** *For all $z_0 \in [1, t]$ and $z \in (1, t)$, we have $H(z) := 2\mu(z) + (z_0 - z)\mu'(z) > 0$.*

Proof. H is affine in z_0. At $z_0 = 1$ it equals $(z-1)\big((z-1)^2 + t - 1\big) > 0$. At $z_0 = t$ it equals $(t-z)\big(t(t-1) - (t-z)^2\big) > 0$. □

Proposition 3.8 (Reduction to the event lemma)**.** *Suppose there is a function $f \in C^2((1, z_0]) \cap C^0([1, z_0])$ with the following properties:*

(a) $f(1) = 0$, $f'(z) = o\big((z-1)^{-1}\big)$ as $z \to 1$, and $f > 0$ on $(1, z_0]$;
(b) $f'(z_0) \geq 0$;
(c) with $s := z_0 - z$ and $h := f'/f$, on $(1, z_0)$,

$$(C) \quad s\,h\,\mu \leq H, \qquad\qquad (D) \quad \tfrac{1}{2}\big(\mu' h + \mu(h' + h^2)\big) \leq s\,(Lh + L'). \tag{3.4}$$

Then every outgoing solution of (3.3) vanishes.

Proof. Cosmological side. On $[z_0, t]$ take $f_+ = 1$. Then $C^{f_+} = -q_1 H \le 0$ by Lemma 3.7, and $D^{f_+} = qL' \le 0$ because $q \ge 0$ there. The limits of the current are computed as in [3, Lemma 4.2]:

$$\lim_{z\to t} J^{f_+}[u_+] = L(t)q(t)|B_t|^2, \qquad J^{f_+}[u_+](z_0) = -\mu(z_0)q_1 \operatorname{Re}(u'\bar{u})(z_0).$$

Since J^{f_+} is non-increasing and $L(t)q(t) > 0$ by Lemma 3.5, we get $\operatorname{Re}(u'\bar{u})(z_0) \le 0$, with equality only if $B_t = 0$.

Event side. On $[1, z_0]$, write $q = -q_1 s$. Then $C^f \le 0$ is equivalent to (C), and $D^f \le 0$ is equivalent to $(\mu f')' \le 2s(Lf)'$, which is (D). By (a), $\lim_{z\to 1} J^f[u_-] = L(1)f(1)q(1)|B_1|^2 = 0$. Monotonicity gives

$$J^f[u_-](z_0) = -\mu(z_0)q_1 f(z_0) \operatorname{Re}(u'\bar{u})(z_0) + \tfrac{1}{2}q_1\mu(z_0)f'(z_0)|u(z_0)|^2 \le 0.$$

So $\operatorname{Re}(u'\bar{u})(z_0) \ge 0$ by (b).

Conclusion. Comparing the two sides forces $B_t = 0$. At $z = t$ the indicial roots differ by a non-integer, because $\operatorname{Im} e_t \ne 0$ in the open window, so the regular-singular argument following [3, (3.6)] gives $u = 0$. □

By Propositions 3.3, 3.4 and 3.8, the real-frequency part of Theorem 1.1 follows from the next result, which only involves the four numbers (t, z_0, A, B).

Theorem 3.9 (Event lemma)**.** *For all $t > 1$, $z_0 \in (1, t)$, $A > 0$ and $B \ge 5.5t - 1.25$, there exists f satisfying (a)–(c) of Proposition 3.8.*

4. Proof of the event lemma

In terms of f, condition (C) reads $f'/f \le h_C := H/(s\mu) = 2/s + \mu'/\mu$. Condition (D) reads $\widetilde{D}f := (\mu f')' - 2s(Lf)' \le 0$. Put $x_0 = z_0 - 1$ and

$$p = \frac{2x_0 L(1)}{t-1}.$$

We distinguish three regimes:

$$\text{R1: } p \le 1; \qquad \text{R2: } p > 1 \text{ and } 2x_0^2\Big(\frac{A}{z_0} + 3\Big) < t - 1; \qquad \text{R3: the rest.}$$

4.1. Regimes R1 and R2: an explicit solution.

Proposition 4.1. *In R1 and R2, the conclusion of Theorem 3.9 holds.*

Proof. Let $\tilde{L} = L$ in R1, and $\tilde{L} = L - \kappa$ with $\kappa = L(1) - \frac{t-1}{2x_0} > 0$ in R2. Let f solve $(\mu f')' = 2s(\tilde{L}f)'$ with $f(1) = 0$, that is,

$$\mu f' = 2s\tilde{L}f + 2\int_1^z \tilde{L}f.$$

At $z = 1$ this is a regular-singular equation, and $f \sim (z-1)^{\tilde{p}}$ with $\tilde{p} = 2x_0\tilde{L}(1)/(t-1) \in (0, 1]$. So (a) holds. In R2 we have $\tilde{L}(z_0) > 0$, since $L(1) - L(z_0) = Ax_0/z_0 + 3x_0$ and this is exactly the R2 condition. Thus $\tilde{L} > 0$ on $[1, z_0]$, so $\mu f' > 0$ and f is positive and increasing; in particular (b) holds.

For (D): $(\mu f')' = 2s(\tilde{L}f)' \le 2s(Lf)'$, because $((L - \tilde{L})f)' = \kappa f' \ge 0$.

For (C): $\tilde{L}$ is decreasing, so

$$s\mu f' \le 2s\tilde{L}(1)f\,(s + z - 1) = 2sx_0\tilde{L}(1)f \le s(t-1)f.$$

It remains to show that $H \ge s(t-1)$. With $z = 1 + x$,

$$H - s(t-1) = x\big[2(1+x)(t-1-x) + s(2t-4-3x)\big].$$

The bracket is affine in s. It is nonnegative at $s = 0$, and at $s = t - 1 - x$ it equals $(t - 1 - x)(2t - 2 - x) \geq 0$. □

4.2. **Regime R3: saturate (C), then solve (D).**

Proposition 4.2. *In R3, the conclusion of Theorem 3.9 holds.*

The construction follows [3, §4.3.3]. Since $h_C = (\log F_C)'$ with

$$F_C := \mu/s^2,$$

we first take $f = F_C$, for which (C) holds with equality. Condition (D) then becomes $d := \widetilde{D}F_C/(2F_C) \leq 0$. A direct computation gives $\widetilde{D}F_C = N/s^4$, where in the variable $S = z_0 - z \in (0, x_0)$

$$N = 6\mu_0^2 - 6\mu_0\mu_0' S + c_2 S^2 + c_3 S^3 + 2(B + 4t - 15z_0 + 4)S^5 + 15S^6. \tag{4.1}$$

Here $\mu_0 = \mu(z_0)$, $\mu_0' = \mu'(z_0)$, and c_2, c_3 are affine in (A, B) with explicit polynomial coefficients in (t, z_0). The coefficient of S^4 vanishes.

Lemma 4.3 (One sign change)**.** *In R3, N has exactly one zero $\hat{S} \in (0, x_0)$. Hence $d < 0$ near $z = 1$, $d > 0$ near z_0, and d changes sign exactly once, at $\hat{z} = z_0 - \hat{S}$.*

Proof. Substitute $S = x_0 v/(1 + v)$, $v \in (0, \infty)$, and clear denominators. Let $T_0, \dots, T_6$ be the coefficients of the resulting sextic in v.

- *The first two coefficients.* $T_0 = 6\mu_0^2 > 0$, and $T_1 = 6z_0(z_0 - t)x_0^2(3z_0^2 + 2z_0 - 4tz_0 - t) > 0$. Both factors of T_1 are negative; the last factor is convex in z_0 and negative at $z_0 = 1$ and $z_0 = t$.
- *The other coefficients.* For $j \geq 2$, each T_j is affine in (A, B) with negative coefficients.
- *The parameter region.* In R3 the admissible (A, B) form the quadrant $\{A \geq a_0,\ B \geq 5.5t - 1.25\}$, where $a_0 = \max\big(0, z_0((t-1)/(2x_0^2) - 3)\big)$. This uses that the line $A + B = 3 + (t - 1)/(2x_0)$ coming from $p > 1$ is redundant, which is a two-variable inequality.
- *Signs on the quadrant.* By the previous two items, it suffices to check the following at the corner of the quadrant and on the triangle where $T_3 \geq 0$:

  $$T_4, T_5, T_6 < 0, \qquad \text{and} \qquad T_3 \geq 0 \Rightarrow T_2 > 0.$$

  These reduce to finitely many polynomial inequalities in (t, z_0). We certify them exactly in the variables $(1/t, \zeta)$, with $\zeta = x_0/(t - 1)$ (Appendix A, item C3).

Thus the sign pattern of $(T_0, \dots, T_6)$ has exactly one change. By Descartes' rule, N has exactly one positive root in v, and hence exactly one root in $S \in (0, x_0)$. □

Proof of Proposition 4.2. Define $f = F_C$ on $(1, \hat{z}]$, and on $[\hat{z}, z_0]$ let f solve $\widetilde{D}f = 0$ with $(f, f')(\hat{z}) = (F_C, F_C')(\hat{z})$. Since $d(\hat{z}) = 0$, f is C^2. Condition (a) holds because $F_C \sim (t - 1)(z - 1)/x_0^2$ as $z \to 1$. Condition (D) holds on $(1, \hat{z}]$ by Lemma 4.3, and with equality on $[\hat{z}, z_0]$.

(C) on $[\hat{z}, z_0]$. Let $p_* = \mu \exp\big(-\int 2sL/\mu\big) > 0$. Then $\widetilde{D}f = (\mu/p_*)(p_* f')' - 2sL'f$. The Wronskian $W = p_*(F_C f' - F_C' f)$ satisfies $W' = -(p_*/\mu) f\, \widetilde{D}F_C \leq 0$ wherever $f > 0$, because $d \geq 0$ on $[\hat{z}, z_0]$. Since $W(\hat{z}) = 0$, we get $f'/f \leq F_C'/F_C = h_C$.

Positivity and the seam. At a zero of f' with $f > 0$, $\mu f'' = 2sL'f < 0$. Moreover $f'(\hat{z}) = H(\hat{z})/\hat{s}^3 > 0$. So it suffices to show $f'(z_0) > 0$.

Suppose not, and let $z_1 \in (\hat{z}, z_0]$ be the first zero of f'. The function $\Phi = \mu f' - 2sLf$ satisfies $\Phi' = 2Lf \geq 2L\hat{f}$ on $[\hat{z}, z_1]$, where $\hat{f} = f(\hat{z})$. This yields $\Psi(z_1) \leq 0$ for

$$\Psi(z) = \hat{H}/\hat{s} - 2\hat{s}\hat{L} + 2\int_{\hat{z}}^{z} L + 2s(z)L(z).$$

But $\Psi' = 2sL' < 0$, so $\Psi(z_1) \geq \Psi(z_0) = \hat{H}/\hat{s} - 2\int_{\hat{z}}^{z_0} s|L'|$. Using $u - \log(1+u) \leq u^2/2$,

$$\int_z^{z_0} s|L'| = A\Big(\frac{z_0}{z} - 1 - \log\frac{z_0}{z}\Big) + \frac{3}{2}s^2 \leq s^2\Big(\frac{A}{2z^2} + \frac{3}{2}\Big),$$

so it suffices that $K(\hat{z}) > 0$, where $K = H - s^3(A/z^2 + 3)$.

Now K is affine and decreasing in A, and N is affine in A and in B with negative coefficients on $(0, x_0)$. Hence the implication

$$K \leq 0 \implies N(S; \max(A_K, 0), 5.5t - 1.25) < 0,$$

where A_K is the value of A at which $K = 0$, implies $K \leq 0 \Rightarrow N < 0$ for all admissible (A, B). The implication is a three-variable polynomial statement, which we certify exactly (Appendix A, item C4). Since $N \geq 0$ on $[0, \hat{S}]$ by Lemma 4.3, it follows that $K(\hat{z}) > 0$. Thus $f' > 0$ on $[\hat{z}, z_0]$; in particular $f > 0$, and (b) holds. □

This completes the proof of Theorem 3.9, and with it Theorem 1.1 for real $\sigma \neq 0$.

5. Zero frequency

Proposition 5.1. *There are no nontrivial outgoing modes at $\sigma = 0$, for $s = \pm 2$.*

Proof. Case $m \neq 0$. Then $w = 0 < w_2$ lies outside the window, and Corollary 3.2 applies. Indeed, $\operatorname{Im}\kappa_1$ and $\operatorname{Im}\kappa_3$ are nonzero because $am \neq 0$.

Case $m = 0$, $s = -2$. The radial equation is $(\Delta_r^{-1}R')' + V\Delta_r^{-2}R = 0$, with $V = -2\Lambda r^2 - 2(1-\alpha) - \lambda$. Put $g = \Delta_r^2 R''''$. The Teukolsky–Starobinsky identity at $K = 0$ shows that g solves the same equation. Moreover g vanishes like Δ_r^2 at both horizons, so the energy identity $\int \Delta_r^{-1}|g'|^2 = \int V\Delta_r^{-2}|g|^2 \leq 0$ gives $g = 0$. Thus R is a cubic polynomial. Inserting a cubic into the equation gives a 4×4 linear system for its coefficients, with determinant $[\lambda^2 + 6(1-\alpha)\lambda + 8 - 4\alpha + 8\alpha^2]^2$. Both roots of the quadratic are negative, and $\lambda \geq 2 - 6\alpha > 0$ at $m = \xi = 0$ by the perturbation bound in the proof of Proposition 3.6. Hence the determinant is positive and $R = 0$.

Case $m = 0$, $s = +2$. The function $\Delta_r^2 R_{+2}$ solves the $s = -2$ equation, so we may repeat the argument.

The two identities used are exact symbolic identities (Appendix A, item C5). □

6. The upper half plane

6.1. **Fredholm theory.** Fix $m \in \mathbb{Z}$. On the subspace $\ker(\partial_{\varphi_*} - im)$, the spectral family $P_b(\sigma)$ of the Teukolsky operator of spin s at parameters $b = (\Lambda, M, a)$ has the following properties. They follow from Vasy's framework [13] with the dynamical input of Petersen–Vasy [10, 9]. As noted in [4, §2.3], a nonvanishing subprincipal part only shifts threshold regularities, and the Teukolsky operator in a horizon-regular frame is a principally scalar wave operator on a line bundle.

- (F1) $P_b(\sigma)$ is an analytic family of Fredholm operators of index 0 on $\operatorname{Im}\sigma > -c$, depending continuously on b. Its non-invertibility set is discrete, and it is the set of quasinormal mode (QNM) frequencies.
- (F2) There is C_0, locally uniform in b, such that every QNM with $\operatorname{Im}\sigma \geq 0$ has $\operatorname{Im}\sigma \leq C_0$. This follows from a spacetime energy estimate [3, §6, item (3)].
- (F3) The set $\{(b, \sigma) : P_b(\sigma) \text{ is not invertible}\}$ is closed. If $P_{b_0}(\sigma_0)$ is not invertible, then for b near b_0 there is a QNM near σ_0 [5, §5].

What is *not* available for Teukolsky in the full subextremal range is the high-energy estimate bounding $|\operatorname{Re}\sigma|$. We obtain it sector by sector from the gauge-fixed Einstein operator, for which it is proved in [4].

6.2. **Metric reconstruction.** We use the Geroch–Held–Penrose (GHP) formalism with $\text{þ}' = \Delta - p\gamma - q\bar{\gamma}$ and $\text{ð}' = \bar{\delta} - p\alpha - q\bar{\beta}$ [2]. Let $\mathcal{O}$ be the GHP Teukolsky operator for Ψ_0, and $\mathcal{O}'$ its prime. On any type D Λ-vacuum the Bianchi identities give $\text{þ}\Psi_2 = 3\rho\Psi_2$, $\text{ð}\Psi_2 = 3\tau\Psi_2$ and their primes; there are no Λ-terms. With $P := \Psi_2^{-4/3}$ (GHP type $(0,0)$), and using the formal adjoint rules $\text{þ}^\dagger = -(\text{þ} - \rho - \bar{\rho})$ and $\text{ð}^\dagger = -(\text{ð} - \tau - \bar{\tau}')$, a direct computation gives

$$\mathcal{O}'^\dagger \circ P = P \circ \mathcal{O}. \tag{6.1}$$

For instance, the radial part is

$$(\text{þ} - \bar{\rho})(\text{þ}' + 3\rho')(PX) = P(\text{þ} - 4\rho - \bar{\rho})(\text{þ}' - \rho')X.$$

Wald's argument [14], which uses only the decoupling identity $\mathcal{S}'\mathcal{E} = \mathcal{O}'\mathcal{T}'$ and the self-adjointness of the linearized Einstein operator $\mathcal{E}$, then gives the following.

Lemma 6.1 (Reconstruction)**.** *If $\mathcal{O}\Psi = 0$, then $h_\Psi := \mathcal{S}'^\dagger(P\Psi)$ solves $\mathcal{E}h_\Psi = 0$. Moreover, the complex conjugate $\tilde{h} := \overline{h_\Psi}$ satisfies $\Psi_4[\tilde{h}] = \frac{1}{2}(\text{þ}')^4(\bar{P}\bar{\Psi})$.*

Proof. The first claim is Wald's argument combined with (6.1). For the second: h_Ψ has components only along $l\otimes l$, $l\otimes m$ and $m\otimes m$. By GHP type counting, $\delta\Psi_4$ involves only the components h_{nn}, $h_{n\bar{m}}$ and $h_{\bar{m}\bar{m}}$. Hence $\Psi_4[h_\Psi] = 0$, and $\Psi_4[\tilde{h}]$ comes from $\tilde{h}_{\bar{m}\bar{m}}$ alone. Its evaluation uses only $\text{þ}'\rho' = \rho'^2$ and contains no Λ-term [14, 7]. □

6.3. **The kernel lemma.** Let $\kappa_2^\sharp = |\Delta_r'(r_2)|/(2\Xi\rho_2)$ be the normalized surface gravity of the cosmological horizon, and define the *cosmological ladder*

$$\mathcal{Z}_m = \{m\Omega_2 + iN\kappa_2^\sharp : \ N \in \tfrac{1}{4}\mathbb{Z}\}.$$

Lemma 6.2 (Kernel lemma)**.** *Let $\operatorname{Im}\sigma \neq 0$ and $\sigma \notin \mathcal{Z}_m$. If $F = e^{-i\sigma t_* + im\varphi_*}v$ is a mode of GHP type $(0,4)$, regular at the future cosmological horizon, with $(\text{þ}')^4F = 0$, then $F = 0$.*

Proof. The operator $\text{þ}'$ differentiates only along n, which has no ∂_θ-component. So for each fixed θ, $(\text{þ}')^4F = 0$ is a fourth-order linear ODE in r, a composition of four first-order operators with a regular singular point at r_2. The n-invariant functions are those of the advanced time $v = t + r_*$, and of $\varphi + \int a/\Delta_r$. The integrating factors coming from the weights $p\gamma + q\bar{\gamma}$ are real powers of Δ_r, with exponents in $\frac{1}{4}\mathbb{Z}$, times analytic functions. Hence every local solution near r_2 is a finite sum of terms $(r_2 - r)^\beta \log^j(r_2 - r)$ times analytic functions, with $\beta \in -\kappa_2 + \frac{1}{4}\mathbb{Z}$.

On the other hand, regularity at the future cosmological horizon means smoothness in the retarded time. By Lemma 2.2 this is the branch $(r_2 - r)^{\kappa_2}C^\infty$. Since the spin weights contribute only real exponents in $\frac{1}{4}\mathbb{Z}$, the two descriptions are compatible for $F \neq 0$ only if $2\kappa_2 \in \frac{1}{4}\mathbb{Z}$. By the definition of κ_2, this means exactly $\sigma \in \mathcal{Z}_m$. The arguments below only use that $\mathcal{Z}_m$ is a discrete subset of the line $\{\operatorname{Re}\sigma = m\Omega_2\}$. Finally, F vanishes identically once it vanishes near r_2, by uniqueness for the ODE. □

Proposition 6.3 (Transfer)**.** *Let Ψ be an outgoing $s = +2$ Teukolsky mode at frequency σ in the sector m, with $\operatorname{Im}\sigma > 0$ and $\sigma \notin \mathcal{Z}_m$. Then $\tilde{h}$ from Lemma 6.1 is a smooth outgoing mode of the linearized Einstein equation, at frequency $-\bar{\sigma}$ in the sector $-m$, with $\Psi_4[\tilde{h}] \neq 0$. The case $s = -2$ is the same with the roles of l and n, and of the two horizons, exchanged: one uses the ingoing gauge and the ladder $\mathcal{Z}_m^e$ of the event horizon.*

Proof. Smoothness. $\mathcal{S}'^\dagger$ is GHP-covariant. Near each future horizon it may therefore be evaluated in a principal tetrad that is smooth there. In such a tetrad, outgoing modes are smooth, so $\tilde{h}$ is smooth.

Nonvanishing of Ψ_4. The function $\bar{P}\bar{\Psi}$ is regular at both horizons, and its frequency $-\bar{\sigma}$ lies off $\mathcal{Z}_{-m} = -\overline{\mathcal{Z}_m}$. By Lemma 6.2, $(\text{þ}')^4(\bar{P}\bar{\Psi}) \neq 0$. □

6.4. **Compactness, sector by sector.**

Lemma 6.4. *Let b_0 be subextremal and $m \in \mathbb{Z}$. There are a neighborhood $U \ni b_0$ and $R_m < \infty$ such that, for all $b \in U$, every Teukolsky QNM of spin ± 2 in the sector m with $\operatorname{Im}\sigma \geq 0$ satisfies $|\sigma| \leq R_m$.*

Proof. Let L_b be the linearized gauge-fixed Einstein operator of [4, (23)], with the constraint damping fixed at b_0. By [4, Remark 2.12, Theorem 2.13], the operators L_b, $b \in U$, form a continuous family that satisfies the hypotheses of [5, §5.1.2] uniformly. The high-energy estimates are therefore uniform, and there is R_0 such that every QNM of L_b with $\operatorname{Im} \geq 0$ has $|\sigma_T| \leq R_0$ for all $b \in U$. Here σ_T is the T-frequency, which differs from the ∂_t-frequency by $m'\Omega_T$ on the sector m'.

Let Ψ be a QNM in the sector m with $\operatorname{Im}\sigma \geq 0$. If $\operatorname{Im}\sigma = 0$ there is nothing to prove, by Theorem 1.1 on the real axis. By (F2), $\operatorname{Im}\sigma \leq C_0$, so the ladder points in the strip $0 < \operatorname{Im}\sigma \leq C_0$ satisfy $|\sigma| \leq |m|\Omega_2 + C_0$, or $|\sigma| \leq |m|\Omega_1 + C_0$ for the event-horizon ladder. If σ is not a ladder point, Proposition 6.3 gives $\tilde{h}$ at frequency $-\bar{\sigma}$ with $\Psi_4[\tilde{h}] \neq 0$.

Next we gauge-fix $\tilde{h}$: put $h' = \tilde{h} + \delta^*\omega$, where ω solves the linearized gauge condition $D\Upsilon(\delta^*\omega) = -D\Upsilon(\tilde{h})$ at frequency $-\bar{\sigma}$. The operator $\omega \mapsto D\Upsilon(\delta^*\omega)$ is a principally scalar wave operator on 1-forms, so it is Fredholm of index 0 by (F1) in its version for 1-forms.

- If it is invertible at $-\bar{\sigma}$, then h' is a QNM of L_b, and $h' \neq 0$ since $\Psi_4[h'] = \Psi_4[\tilde{h}] \neq 0$.
- If it is not invertible, there is $\omega \neq 0$ with $D\Upsilon(\delta^*\omega) = 0$. Then $L_b(\delta^*\omega) = 0$, and $\delta^*\omega \neq 0$ because KdS has no Killing modes with $\operatorname{Im} \neq 0$. So $\delta^*\omega$ is a QNM of L_b at frequency $-\bar{\sigma}$.

In either case L_b has a QNM at T-frequency $-\bar{\sigma} - (-m)\Omega_T$, so $|\sigma| \leq R_0 + |m|\Omega_T$. For $s = -2$ the event-horizon ladder enters instead. Hence we can take $R_m = \max(R_0 + |m|\Omega_T,\ |m|\Omega_1 + C_0,\ |m|\Omega_2 + C_0)$. □

6.5. **The base case.**

Lemma 6.5 (Schwarzschild–de Sitter)**.** *There is a dense set of SdS parameters with $\Lambda M^2 < 1/9$ at which there are no Teukolsky QNMs of spin ± 2 with $\operatorname{Im}\sigma > 0$.*

Proof. Let $a = 0$ and $\operatorname{Im}\sigma > 0$. Suppose first that σ is off the ladder. Then Proposition 6.3 produces a linearized Einstein mode with $\operatorname{Im} > 0$ and $\Psi_4 \neq 0$. By [5, §7], every such mode on SdS is pure gauge, and pure gauge modes have $\Psi_4 = 0$. This is a contradiction.

At $a = 0$ the ladder is $\{iN\kappa_2^\sharp\}$ for every m, and the angular operator does not depend on σ. So modes separate into spin-weighted spherical harmonics for every $\sigma \in \mathbb{C}$. On a separated solution, $(\text{þ}')^4$ acts as the radial Teukolsky–Starobinsky operator. By the Teukolsky–Starobinsky identities [1, 11], it annihilates a solution only if $\mathcal{C}_\ell(\sigma) = 0$. Here $\mathcal{C}_\ell$ is a polynomial in σ, continuous in ΛM^2, whose Schwarzschild limit has only the zeros $\pm i(\ell-1)\ell(\ell+1)(\ell+2)/(12M) \neq 0$. Since $\kappa_2^\sharp \to 0$ as $\Lambda \to 0$, for each (ℓ, N) the coincidence $iN\kappa_2^\sharp \in \mathcal{C}_\ell^{-1}(0)$ holds only on a closed nowhere dense set of ΛM^2. Off the countable union of these sets, Lemma 6.2 can be replaced by this nonvanishing, and the argument of the first paragraph applies. □

6.6. **Conclusion.**

Proof of Theorem 1.1 for $\operatorname{Im}\sigma > 0$. Fix a subextremal b_1 and $m \in \mathbb{Z}$. Choose an SdS point b_* as in Lemma 6.5, and a continuous path $p : [0,1] \to \{\text{subextremal parameters}\}$ with $p(0) = b_*$ and $p(1) = b_1$. For example, scale r_0 to 0 with r_1, r_2 fixed, as in [3, §6], and then move within the SdS family. Let S be the set of $s \in [0,1]$ for which the sector m has a QNM with $\operatorname{Im}\sigma > 0$.

- S is open, by (F3).

- S is closed. Let $s_j \to s_0$ with $s_j \in S$ and QNMs σ_j. By Lemma 6.4 and (F2), the σ_j lie in the compact set $\{0 \le \operatorname{Im}\sigma \le C_0,\ |\sigma| \le R_m\}$. A subsequence converges to some σ_0, which is a QNM at $p(s_0)$ by (F3). Now $\sigma_0 \neq 0$, because $P(0)$ is invertible by Proposition 5.1 and invertibility is open. And $\sigma_0 \notin \mathbb{R}$ by §4 and Corollary 3.2. Hence $\operatorname{Im}\sigma_0 > 0$.
- $0 \notin S$, by Lemma 6.5.

Hence $S = \emptyset$. Generalized modes reduce to modes, since their top coefficient is a mode. □

7. Ungauged linearized Einstein mode stability

Throughout this section $a \neq 0$, and $\Omega_T = a/(r_2^2 + a^2)$, so that $T = \partial_t + \Omega_T \partial_\varphi$ in Boyer–Lindquist coordinates. Let h be as in Theorem 1.2. Its Fourier components in φ_* are generalized $(\partial_t, \partial_\varphi)$-modes of type $(\sigma + m\Omega_T, m)$, whose imaginary part is $\operatorname{Im}\sigma \ge 0$. The linearized Teukolsky scalars $\delta\Psi_0$ and $\delta\Psi_4$ are gauge and tetrad invariant on type D, and they are outgoing generalized Teukolsky modes. By Theorem 1.1, applied to each Fourier component of h and of $\bar{h}$, we get

$$\delta\Psi_0 = \delta\Psi_4 = 0.$$

7.1. Linearized Goldberg–Sachs.

Lemma 7.1. *After an infinitesimal null rotation of the tetrad, $\delta\Psi_1 = \delta\Psi_3 = 0$, and then $\delta\kappa = \delta\sigma = \delta\kappa' = \delta\sigma' = 0$.*

Proof. To first order, a null rotation about n with parameter c shifts $\delta\Psi_1$ by $3c\Psi_2$ and fixes $\delta\Psi_0$ and $\delta\Psi_2$; the analogous statement holds about l. Since $\Psi_2 \neq 0$, the choice is algebraic. The Λ-vacuum Bianchi identities contain no Λ-terms. Linearizing

$$\text{Þ}\Psi_1 - \eth'\Psi_0 = -\tau'\Psi_0 + 4\rho\Psi_1 - 3\kappa\Psi_2, \qquad \text{Þ}'\Psi_0 - \eth\Psi_1 = \rho'\Psi_0 - 4\tau\Psi_1 + 3\sigma\Psi_2$$

gives $\delta\kappa = \delta\sigma = 0$; priming gives the rest. □

All perturbed quantities below are local linear functions of h. They are boost-spin invariant and commute with $\mathcal{L}_{\partial_t}$ and $\mathcal{L}_{\partial_\varphi}$.

7.2. The Killing-spinor field.

On a type D Λ-vacuum with $\kappa = \sigma = \kappa' = \sigma' = 0$, the following identities hold, together with their primes:

$$\text{Þ}\rho = \rho^2, \quad \eth\rho = (\rho - \bar{\rho})\tau, \quad \text{Þ}\tau = (\tau - \bar{\tau}')\rho, \quad \eth\tau = \tau^2,$$
$$\text{Þ}\rho' - \text{Þ}'\rho = \tau\bar{\tau} - \tau'\bar{\tau}', \quad \eth\tau' - \eth'\tau = \rho\bar{\rho}' - \bar{\rho}\rho', \quad \text{Þ}\tau' = \eth'\rho, \quad \text{Þ}'\tau = \eth\rho'.$$

They follow from the GHP Ricci identities and commutators. We verified them exactly on KdS, and on the accelerating NUT Plebański–Demiański metric, which has $\tau \neq -\bar{\tau}'$ (Appendix A, item C6).

Let $\phi = \Psi_2^{-1/3}$ (normalized so that $\phi = r - ia\cos\theta$ on KdS), and put

$$A = \phi\rho, \quad B = -\phi\rho', \quad C = \phi\tau, \quad D = -\phi\tau', \qquad \xi = Bl + An - Dm - C\bar{m}.$$

Lemma 7.2. *The ten Killing equations for ξ are algebraic consequences of the identities above. On KdS, $\xi = -\Xi\,\partial_t$.*

Linearizing Lemma 7.2 gives $\mathcal{L}_\xi h + \mathcal{L}_{\delta\xi} g = 0$, where $\delta\xi = \delta\xi[h]$.

Lemma 7.3 (Reality)**.** *Put $J[h] = \big(\delta\xi[h] - \overline{\delta\xi[\bar{h}]}\big)/(2i)$. Then $\mathcal{L}_{J[h]} g = 0$. If $(\sigma, m) \neq (0,0)$, then $J[h] = 0$.*

Proof. For real h, $J[h] = \operatorname{Im}\delta\xi$, and the imaginary part of the linearized Killing equation reads $\mathcal{L}_J g = 0$; by complex linearity this holds for all h.

The local Killing algebra of KdS with $a \neq 0$ is $\operatorname{span}\{\partial_t, \partial_\varphi\}$ on every open set. Indeed, a Killing field preserves Ψ_2, hence r and $a\cos\theta$. It is therefore tangent to the (t,φ)-orbits, and the isotropy is trivial because it must fix ∇r, $\nabla(a\cos\theta)$ and the principal null directions.

Moreover, J is complex linear and commutes with $\mathcal{L}_{\partial_t}$ and $\mathcal{L}_{\partial_\varphi}$. For a mode of type $(\sigma, m) \neq (0,0)$, it is therefore a constant combination of $\partial_t, \partial_\varphi$ of that type, hence zero. □

Remark 7.4. The same argument shows $J[h] = 0$ in two further cases. If $(\mathcal{L}_T + i\sigma)h = 0$ with $\sigma \neq 0$, then $\mathcal{L}_T J[h] = -i\sigma J[h]$, while $\mathcal{L}_T$ annihilates every Killing field of KdS. If $\mathcal{L}_T h = 0$ and h has vanishing φ-average, then $J[h]$ is a constant combination of $\partial_t, \partial_\varphi$, hence equal to its own φ-average, which is J of the φ-average of h, namely 0.

Consequently $\phi\rho = \bar\phi\bar\rho$, $\phi\rho' = \bar\phi\bar\rho'$ and $\phi\tau = -\bar\phi\bar\tau'$ hold to first order.

7.3. **Carter's Killing field.** Let $R = \operatorname{Re}\phi$ and $I = \operatorname{Im}\phi$. These are the Carter coordinates r and $-a\cos\theta$. Reality gives

$$\text{Þ}R = -A, \quad \text{Þ}'R = B, \quad \text{ð}R = 0, \quad \text{Þ}I = \text{Þ}'I = 0, \quad \text{ð}I = iC.$$

Define

$$\zeta = I^2(Bl + An) + R^2(Dm + C\bar m).$$

Lemma 7.5. *To first order, ζ is Killing. On KdS, $\zeta = -\Xi(a^2\partial_t + a\partial_\varphi)$.*

Proof. The ll, nn, mm and $\bar m\bar m$ components are immediate from those of ξ.

- The ln component equals $-|\phi|^2\phi(\tau\bar\tau - \tau'\bar\tau') = 0$, since by reality $\tau\bar\tau = \tau'\bar\tau'$.
- The $m\bar m$ component equals $|\phi|^2\phi(\rho\bar\rho' - \rho'\bar\rho) = 0$, again by reality.
- The lm component equals $2AC\phi + |\phi|^2(\text{ð}A + \bar\tau' A) = 0$, using $\text{ð}A = -\phi\bar\rho\tau$.
- The nm component equals $-2BC\bar\phi + |\phi|^2(\text{ð}B + \tau B)$. Differentiating the first-order reality relation $\phi\rho' = \bar\phi\bar\rho'$ along m gives $\text{ð}B + \tau B = 2\tau B$, and the sum vanishes.

The remaining components are complex conjugates. The value on KdS is an exact computation (item C6). □

7.4. **Pure gauge modes.** On KdS, $\partial_t = -\xi/\Xi$ and $\partial_\varphi = (a^2\xi - \zeta)/(\Xi a)$. Hence there are constants α_T, β_T and $\alpha_\varphi, \beta_\varphi$ with

$$T = \alpha_T\xi + \beta_T\zeta, \qquad \partial_\varphi = \alpha_\varphi\xi + \beta_\varphi\zeta.$$

Whenever $J[h] = 0$, Lemmas 7.2 and 7.5 give $\mathcal{L}_\xi h = -\mathcal{L}_{\delta\xi} g$ and $\mathcal{L}_\zeta h = -\mathcal{L}_{\delta\zeta} g$, and therefore

$$\mathcal{L}_T h = -\mathcal{L}_{Y_T} g, \qquad \mathcal{L}_{\partial_\varphi} h = -\mathcal{L}_{Y_\varphi} g, \qquad Y_\bullet = \alpha_\bullet\delta\xi + \beta_\bullet\delta\zeta. \tag{7.1}$$

The vector fields Y_T and Y_φ are local linear expressions in h. They are smooth across the horizons: ξ, ζ, $\delta\xi$ and $\delta\zeta$ are boost-spin invariant constructions from the perturbed Weyl spinor, so they can be computed in a horizon-regular principal tetrad, and $\Psi_2 \neq 0$ on $r > 0$.

Proposition 7.6. *(i) If $(\mathcal{L}_T + i\sigma)h = 0$ with $\sigma \neq 0$, then $h = \mathcal{L}_X g$ with $X = Y_T/(i\sigma)$.*

(ii) If $\mathcal{L}_T h = 0$ and h has vanishing φ-average, then $h = \mathcal{L}_X g$, where X is the unique φ-antiderivative of $-Y_\varphi$ with vanishing φ-average.

In both cases X is smooth, and $(\mathcal{L}_T + i\sigma)X = 0$.

Proof. In both cases $J[h] = 0$ by Remark 7.4, so (7.1) holds.

(i) Here $-i\sigma h = \mathcal{L}_T h = -\mathcal{L}_{Y_T} g$.

(ii) The field Y_φ has vanishing φ-average, because Y_φ commutes with rotations and h has vanishing average. So its mean-zero φ-antiderivative X (with $[\partial_\varphi, X] = -Y_\varphi$) is smooth. Then

$\mathcal{L}_{\partial\varphi}(h-\mathcal{L}_X g) = -\mathcal{L}_{Y_\varphi} g + \mathcal{L}_{Y_\varphi} g = 0$. So $h - \mathcal{L}_X g$ is axisymmetric with vanishing average, hence zero. □

7.5. **The stationary axisymmetric sector.**

Proposition 7.7. *If $\mathcal{L}_{\partial_t} h = \mathcal{L}_{\partial_\varphi} h = 0$ and $\delta\Psi_0 = \delta\Psi_4 = 0$, then $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$, with X affine in (t, φ).*

Proof. Step 1: restoring reality. Here $J[h] = A'\partial_t + B'\partial_\varphi$ need not vanish. A constant phase change of ϕ removes A'. Subtracting a multiple of the tangent to the Plebański–Demiański family in the direction of the acceleration removes B': this is the only Plebański–Demiański tangent that makes ξ complex, and its J has nonzero ∂_φ-part (exact nonlinear check, item C6). After this, reality holds, and ξ and ζ are first-order Killing fields. They commute, since $\delta\xi$ and $\delta\zeta$ are background Killing fields.

Step 2: Carter's ansatz. One has $\nabla R \perp \nabla I$. Put $\Delta_R := -2AB|\phi|^2$ and $\Delta_I := -2CD|\phi|^2$. Then Δ_R depends on R alone and Δ_I on I alone, since $\eth(AB|\phi|^2) = 0$ by reality and $\xi(\Delta_R) = 0$. All the inner products of $\xi, \zeta, \nabla R$ and ∇I are functions of $(R, I, \Delta_R, \Delta_I)$. In coordinates adapted to (ξ, ζ, R, I), the perturbed metric is therefore, to first order, Carter's separable ansatz

$$\frac{S}{\Delta_R} dR^2 + \frac{S}{\Delta_I} dI^2 + \frac{\Delta_I (d\tilde{t} + R^2 d\psi)^2 - \Delta_R (d\tilde{t} - I^2 d\psi)^2}{S}, \qquad S = R^2 + I^2.$$

Step 3: the field equations. On this ansatz the Λ-vacuum equations are linear in (Δ_R, Δ_I), with general solution

$$\Delta_R = -\tfrac{\Lambda}{3} R^4 + \varepsilon R^2 - 2\mathfrak{m} R + \gamma, \qquad \Delta_I = -\tfrac{\Lambda}{3} I^4 - \varepsilon I^2 + 2\mathfrak{n} I + \gamma$$

(exact computation, item C7). Modulo the rescaling gauge, the variations are δM, δa and the NUT variation $\delta\mathfrak{n}$, together with the acceleration removed in Step 1.

Step 4: axis regularity. Elementary flatness at the two axes gives two linear conditions on $(\delta\mathfrak{n}, \delta\alpha_{\mathrm{acc}})$ with determinant $-64 M p_0^4 / c \neq 0$, so both vanish. The gauge vector X is the generator of the change from the background to the perturbed adapted coordinates, together with the rescalings of ξ and ζ.

Step 5: smoothness of X. Build the adapted coordinates in the regular form, using the advanced time $\tilde{v}$ near the event horizon and the retarded time $\tilde{u}$ near the cosmological horizon, both defined from $(\tilde{\xi}, \tilde{\zeta}, \tilde{R}, \tilde{I})$ and the principal null directions of the perturbed metric, exactly as t_* is built from Boyer–Lindquist coordinates. These are natural constructions from the perturbed geometry, and they are smooth across the respective future horizons. Comparing them with the corresponding background coordinates yields $X = c\, t_* \partial_{t_*} + X_0$ with X_0 stationary, axisymmetric and smooth. Here $\partial_M g$ and $\partial_a g$ denote the parameter derivatives of the KdS family in the regular coordinates $(t_*, r, \theta, \varphi_*)$, as in [4]. □

7.6. **Generalized modes and proof of Theorem 1.2.**

Proof of Theorem 1.2. Part (1). Let $(\mathcal{L}_T + i\sigma)^k h = 0$ with $\sigma \neq 0$. We argue by induction on k; the case $k = 1$ is Proposition 7.6(i). For $k \geq 2$, the top coefficient $h_1 = (\mathcal{L}_T + i\sigma)^{k-1} h$ is a mode, so $h_1 = \mathcal{L}_{X_1} g$ with $(\mathcal{L}_T + i\sigma) X_1 = 0$. Put $Y = t_*^{k-1} X_1 / (k-1)!$, which is smooth since t_* is regular. Then $(\mathcal{L}_T + i\sigma)^{k-1} Y = X_1$, so $h - \mathcal{L}_Y g$ has order $k - 1$.

Part (2). Let $(\mathcal{L}_T)^k h = 0$, and split $h = h_0 + h_\perp$ into its φ-average and the rest. The same induction, using Proposition 7.6(ii), shows that $h_\perp$ is pure gauge. The average h_0 satisfies $\mathcal{L}_{\partial_\varphi} h_0 = 0$ and $(\mathcal{L}_{\partial_t})^k h_0 = 0$. For $k = 1$, Proposition 7.7 applies. For $k \geq 2$:

- Suppose $\mathcal{L}_{\partial_t} h_0 = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$. The Iyer–Wald charges δH_K, for $K \in \{\partial_t, \zeta\}$, are independent of the 2-sphere. They vanish on $\mathcal{L}_X g$ for every smooth X: from $\Theta(\mathcal{L}_X g) = X \cdot L + dQ_X$ and naturality of Q, one gets $\delta H_K[\mathcal{L}_X g] = -\oint (Q_{[X,K]} + Q_{[K,X]}) = 0$.

Applying them on the spheres $\{t_* = \tau\}$ forces $\delta H_K[c_M \partial_M g + c_a \partial_a g] = 0$. Since $(E, J) = (M/\Xi^2, aM/\Xi^2)$ has Jacobian $M/\Xi^4 \neq 0$ with respect to (M, a), it follows that $c_M = c_a = 0$. Then $h_0 - \mathcal{L}_{t_* X} g$ has lower order, and the induction proceeds.

□

Appendix A. Computer-assisted steps

All computations use exact arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(i)$, except item C2, which uses rigorous interval arithmetic. The code is in the repository accompanying this paper; the script names are given in brackets.

C1. *Dual potential* [`step39_c1_symbolic.py`, `step89_c1_kappa.py`]. After Vieta elimination, the numerators of $n_1 - (\lambda C_0 + c_1)$ and $n_2 + 3$ are divisible by $s^2 - 4$ as polynomials.

C2. *Angular bound* [`step47_angular_proof.py`]. Interval evaluation of the two-level lower bound, with exact Beta moments, over $(\alpha, \xi) \in [0, 0.0718] \times [0, |m|]$ for $|m| \in \{1, 2\}$ and $s = \pm 2$.

C3. *One sign change* [`step119_build.py`, `cert2/p1chart.py`, `cert2/run_P1_all.sh`]. Seventeen polynomials in $(1/t, \zeta)$ of degree at most 11. Positivity is certified by integer Bernstein-basis branch and bound on four charts: two rectangles and two blow-up charts at the corner $(1/t, \zeta) = (0, 0)$, after dividing out factors that are positive on the open box. All eight runs certify.

C4. *Seam inequality* [`step121_build.py`, `cert2/p2cert.py`, `cert2/run_P2_all.sh`]. Four polynomials in $(1/t, \zeta, S/x_0)$ of degree at most 7, on nineteen charts covering $[0, 1]^3$, including blow-ups of two degenerate edges. All certify.

C5. *Zero frequency* [`step125_sigma0_check.py`]. The Teukolsky–Starobinsky identity at $K = 0$ for the $s = -2$ equation, and the $4{\times}4$ determinant. For $s = +2$ one substitutes $R_{+2} = \Delta_r^{-2} F$: then $\Delta_r^{-2}(\Delta_r^3 R'_{+2})' = \Delta_r^{-2}(\Delta_r F'' - \Delta'_r F' - 2\Delta''_r F)$, with $\Delta''_r = -4\Lambda r^2 + 2(1-\alpha)$. This matches the r^2-terms of the two potentials, $-10\Lambda r^2 + 8\Lambda r^2 = -2\Lambda r^2$, and the constants are absorbed by the relation between the spin ± 2 angular eigenvalues at $\sigma = 0$.

C6. *Type D identities* [`step92_ghp_check.py`, `step93_*.py`, `step95_pd_xi.py`, `step96_zeta.py`, `step97_r5_pd_check.py`]. These are checks of identities that are already derived by hand; the derivations do not depend on them.

C7. *Carter ansatz* [`step102_carter_linear.py`]. The Ricci tensor of the ansatz with symbolic $\Delta_R(R)$ and $\Delta_I(I)$, and the solution of the resulting linear equations.

The Bernstein enclosure routine was tested against random polynomials and random subdivisions [`cert2/test_bern.py`].

Statement on the use of AI

In preparing this work I used Claude Opus 5.5 (Anthropic) to develop and check proofs, write and run the symbolic and certificate code, search and check the literature, and draft and edit the text. The proofs were also checked in several adversarial review passes, each run by an AI model. These passes found errors, including invalid earlier versions of the large-A regime of the event lemma and of the passage to the upper half plane, and every error found was corrected. Numerical experiments were used only to guide the search for proofs, and no result depends on them. No human expert has yet checked the argument. I have reviewed and edited the content and take full responsibility for it.

Data availability

The code for the computer-assisted steps is available from the author on request.

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Purdue University, West Lafayette, IN, USA

Email address: jpadhi@purdue.edu